Reconstruction of objects from grasps with latent diffusion modeling

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***Abstract*—The goal of this project is to design and train a latent diffusion model that takes geometric information from a anthropomorphic robotics hand grasping an object as an conditional input and reconstructs said object based on the information provided by the input. Research questions of this project are, how much geometric information is encoded in the grasp of a robotic hand and how can this information be used to reconstruct the grasped object. Which latent space encoding and variational autoencoder architecture achieves the best compromise between preserving the information of the original input data and accelerating the training of the model. Which diffusion model architecture offers the best reconstruction quality. Which conditioning technique is best suited for guiding the diffusion process to generate object shapes based on the grasp input. How should the training data be preprocessed or enhanced to improve the training performance of the latent diffusion model.**

Keywords—Latent Space, Diffusion, 3d-object reconstruction, Generative modeling, Humanoid robotics

I. Introduction

Grasping objects is one of the most important tasks in robotics. Recent data-driven approaches tried to solve this task by synthesizing grasps with generative artificial intelligence like variational autoencoders (VAE) or diffusion models from the geometry of the object as an input [1–3]. While being effective in generating a diverse set of possible grasps, these models also generate many invalid grasps and often rely on optimization techniques or discriminator networks to detect invalid grasps. This problem arises from the incomplete geometric representation of the input, because only the front of the object is seen by the robot. To improve the success rate of valid grasp generation, recent works investigated neural shape completion techniques from partial geometric input [4, 5]. These works rely on vision-based systems to provide partial geometric information. But for industrial applications robots must often work under bad lighting conditions or in small, occluded workspaces, where vision cannot be provided. To resolve this issue, other works explored the idea of reconstructing object shapes from tactile sensing [6, 7]. These works use tactile sensors that generate a pressure image of the contact surface from which they extract features with transformer or convolution networks. We are interested in the task of shape completion based on the joint angles of the robot hand, when it is grasping the object. So, we focus on the geometric information that is inherently coded into the hand-object interaction without the reliance on tactile sensors. The closest work that investigated this idea is UGG from [8]. In UGG the authors introduced a unified latent space where object O and grasp G were modeled as joint probability P(G, O). The authors showed that this latent space allows for training both the conditional generation of objects and grasps. While this approach is interesting, we want to develop a direct strategy by directly training the diffusion model to learn the conditional probability of generating objects from a given grasp P(O|G) and compare our results with UGG.

II. Technical Outline

Our main idea is inspired by works like [9, 10] who showed that the DDPM [11] process can be performed in the latent space to accelerate the training, while retaining state of the art generation quality. As we are dealing with high-dimensional 3d data and have limited computing resources, latent diffusion will be key in giving us good generation quality for reasonable computational costs. To further reduce the computational costs, we decided to represent the objects as point clouds. Recent works have proven that the training and reconstruction of point clouds with diffusion models achieve very good results [8, 12–14]. For latent diffusion, the VAE is of great importance as it generates the latent code to perform the diffusion on. For our work, we would like to experiment with different VAEs. Pointnet++ and PVCNN [15, 16] are popular and easy to use autoencoders that generate global shape latent. LION [12], a more advanced architecture constructs both a global shape latent and a point structure latent that better retains the geometrical structure of the original point cloud, thus providing the diffusion model with more geometric information. Then the diffusion model must be considered. Traditional latent diffusion models leverage a U-Net Design that relies on convolution layers [9, 10], but there was also much work done on diffusion transformers for point cloud generation recently [7, 13, 14]. These works showed that transformers are very good in preserving the shape of the original object by tokenizing the object point cloud and using positional encoding on the tokens. We will also investigate different input modalities besides the joint angles of the robot hand, like for instance a point cloud representation of the robot's hand and its effect on reconstruction quality and training time. Different conditioning methods like simple concatenation, FILM or cross-attention will also be evaluated [13, 17]. But for the start of the project, we will keep the model as simple as possible to have a solid baseline that can be improved upon. For the first two weeks we will explore the provided training data and decide which augmentations should be used if needed. Then we will start with a simple latent diffusion architecture that is unconditionally trained to reconstruct point clouds. For this, we will rely on pretrained VAE first and test if this yields good results. This simple testing in the early stages also serves to compare U-Net based and transformer-based diffusion models. Based on the test results, we are going to choose a model architecture and iteratively add more features like grasp conditioning to develop our model.

III. References

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